

Weekly Practical Work 01 – 17/02/2026

Deep Learning Intro and Notion of Tensors

Objectives. The main objective of this Practical Work (PW) is to bring you on track for the **Deep Learning** class with introductory notions. It will include tutorials and exercises to get familiar with Python (if needed), numpy and Pytorch tensors.

Hand-In of PW. The weekly practical works are **optional**. You may submit your solutions to Moodle in which case we will provide you a short feedback. Submissions are in the form of a **zip archive** including your notebooks. Please submit only 1 archive per group : elect 1 student in your group in charge of the submission and indicate in the first cell who are the members of the group. Available submission dates will be indicated on Moodle for each PW.

Exercise 1 Numpy in a Nutshell

If you are not familiar with `numpy`, or if you want a refresh on the manipulation of arrays, read the content of the Jupyter notebook `numpy-tutorial-stud.ipynb` that you will find on Moodle. Pay special attention to the *broadcasting* section that allows to gain significant speedup when processing large numpy arrays. It is usually more efficient to use *broadcasting* instead of `for` loops in your code. At the end of the tutorial, we propose you some manipulations of images stored by arrays to obtain a result as in [1](#).

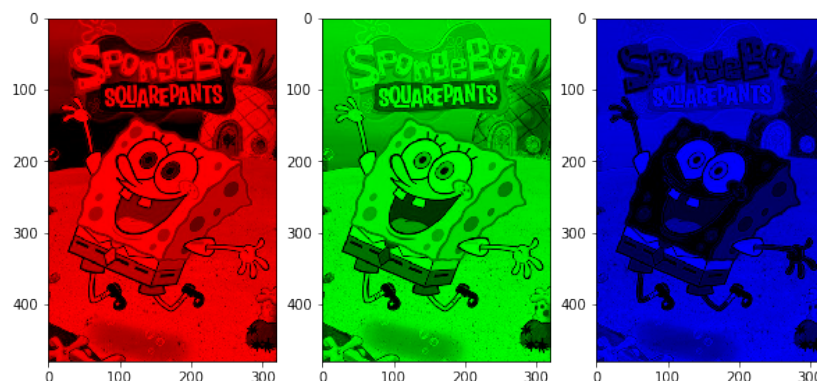


FIGURE 1 – Tri-color planes of Sponge Bob image.

Exercise 2 First hands on Pytorch tensors

- Read the python notebook `tensors-in-practice.ipynb` showing examples of tensor manipulation.
- Do the exercises on tensor manipulation in `tensor_exercices_students.ipynb`

Optional readings and listenings :

- F. Fleuret class 1.4 *Tensor basics and linear regression* and class 1.5 *High dimension tensors* on <https://fleuret.org/dlc/>
- Pytorch tutorial on https://pytorch.org/tutorials/beginner/basics/tensorqs_tutorial.html and get familiar with basic tensor manipulations.

Exercise 3 Tensors – Linear regression

Start from the notebook `linear-regression-stud.ipynb` available on Moodle. We will use a small data set of apartment rent prices, composed of N samples. The objective is to build a linear model to predict the rent price from the surface (m2) of the apartment. By linear model we mean here the equation of a line.

We are then in the case of a monovariale linear regression where we want to discover the parameters θ of a linear model defined with

$$h_{\theta}(\mathbf{x}) = \theta_0 + \theta_1 x \quad (1)$$

The “best” θ ’s are the ones that will minimise the squared difference between the gotten outputs and the *target* outputs defined with

$$J(\theta) = \frac{1}{2N} \sum_{n=1}^N (h_{\theta}(\mathbf{x}_n) - y_n)^2 \quad (2)$$

Equation 2 is a **loss** function, in this case the Mean Squared Error or MSE loss. A closed form solution to this problem is given by the following normal equation :

$$\theta = (X^T X)^{-1} X^T \vec{y} \quad (3)$$

With X and $\vec{y}$ defined as

$$X = \begin{pmatrix} 1 & x_1 \\ 1 & \\ 1 & \vdots \\ 1 & \\ 1 & x_N \end{pmatrix} \quad (4)$$

$$\vec{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix} \quad (5)$$

- a) Following the instructions in the notebook, implement Equation 3 assuming that x is the living area and y is the renting price. Use `numpy` for the vector operations. Plot the computed line on top of the scatter plot.
- b) Compute the overall cost value according to Equation 2.

From the numpy solution, provide a solution to the problem using pytorch.

Exercise 4 Optional Readings

- Read the first chapter of Murphy’s book “Machine Learning”. You can find the pdf on Moodle.
- Read the first chapter of Prince’s book “Understanding Deep Learning” - [online version](#).

Build your own summary of these chapters by doing a taxonomy of machine learning problems.

Exercise 5 Review Questions

Insert your answers in a Python notebook and submit it with the other notebooks.

1) Supervised vs. unsupervised systems

Of the following examples, which one would you address using a supervised or an unsupervised learning algorithm? Give some explanations for your answers.

- a) Given email labeled as spam/not spam, learn a **spam filter**.
- b) Given a set of news articles found on the web, group them into sets of **related articles**.
- c) Given a database of customer data, automatically discover **market segments** and group customers into different market segments.
- d) Given a dataset of patients diagnosed as either having **glaucoma** or not, learn to classify new patients as having glaucoma or not.

2) Classification vs. regression systems

Can we transform a regression problem into a classification problem? What would be the benefits of doing so?

3) Numpy arrays

- a) What are the main differences between a Python *list* and a numpy *array*?
- b) Can we mix different types of data in a numpy array?
- c) Define what is the *rank* in a numpy array.
- d) If we load a gray-level image in a numpy array, what is the rank of this array? Explain the different dimensions. Give a potential value of the **shape** attribute for a square images.
- e) Regarding the previous point, what would be the difference with a color image?